11 Documenting APIs with OpenAPI

This chapter covers

- Understanding OpenAPI and seeing why it's useful
- Adding an OpenAPI description to your app
- Improving your OpenAPI descriptions by adding metadata to endpoints
- Generating a C# client from your OpenAPI description

In this chapter I introduce the OpenAPI specification for describing RESTful APIs, demonstrate how to use OpenAPI to describe a minimal API application, and discuss some of the reasons you might want to do so.

In section 11.1 you'll learn about the OpenAPI specification itself and where it fits in to an ASP.NET Core application. You'll learn about the libraries you can use to enable OpenAPI documentation generation in your app and how to expose the document using middleware.

Once you have an OpenAPI document, you'll see how to do something useful with it in section 11.2, where we add Swagger UI to your app. Swagger UI uses your app's OpenAPI document to generate a UI for testing and inspecting the endpoints in your app, which can be especially useful for local testing.

After seeing your app described in Swagger UI, it's time to head back to the code in section 11.3. OpenAPI and Swagger UI need rich metadata about your endpoints to provide the best functionality, so we look at some of the basic metadata you can add to your endpoints.

In section 11.4 you'll learn about one of the best tooling features that comes from creating an OpenAPI description of your app: automatically generated clients. Using a third-party library called NSwag, you'll learn how to automatically generate C# code and classes for interacting with your API based on the OpenAPI